

RODRIGO GOMES DE ARAÚJO

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SUMMARY

MSc Computer Science student (graduating **July 2027**) focused on systems where **correctness and robustness** are key, such as protocols, consensus and infrastructure people actively rely on. Built a **proof-of-work blockchain** in Rust and is currently extending session types to **enforce correct usage of cryptographic resources** at compile-time, with thesis completion expected in July 2027. Works well **autonomously** and **asynchronously** and emphasizes **testing, documentation and API** in the development process. Comfortable working remotely and async. Based in Porto, Portugal.

EDUCATION

Master's <i>Computer Science</i>	Sep 2025 – Present <i>University of Porto (FCUP)</i>
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Bachelor's <i>Computer Science and Engineering</i>	Sep 2022 – Jul 2025 <i>University of Porto (FCUP)</i>
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RESEARCH

CaST (Cryptographically-aware Session Types) <i>Master's Thesis</i>	Jul 2026 – Present <i>FreeST, Haskell, Type Systems, Applied Cryptography</i>
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- Extending the expressiveness of session types to capture **cryptographic resource usage**, so that its misuse in protocols is detected and rejected at compile-time.
- **Resource tracking composes across nested subprotocols** of arbitrary depth and is preserved under sequential composition and recursion.
- Formalizing the extension and providing a **soundness proof**, integrating it into the FreeST compiler and applying it to smart contracts and protocol specification.

PROJECTS

PoW Blockchain for Decentralized Auctions | Rust, Tokio, LibP2P, Dalek, Docker

- Extensive **research on the state of the art of Bitcoin**, Ethereum and Solana and investigation regarding the security and resilience to failure of these systems.
- Developed a DHT overlay based on Kademlia for **decentralized network construction**.
- PoW Blockchain with **Blake2b hashing**, **Ed25519 signatures** and **Merkle Trees** for transaction validation.
- **Client for auctions** using the blockchain layer as a public ledger.
- Docker **adversarial network simulation** with clients and bootstrap, well behaved and malicious nodes.
- Developed as Data Systems Security final project and graded 19.1/20.

Real-Time Modular Synthesizer | Rust, Tokio, Rodio, Ratatui, SQLite

- **Real-time audio engine** isolated from the UI and control logic, communicating through **lock-free atomics and message passing**. The audio path never locks, allocates or blocks on I/O.
- **Modular**, leveraging Rust's type system. The audio DSP graph is built at runtime and instantiated per note, so voices run **independently** and the sound can be **reconfigured in real-time** without interruption.

AWARDS

2nd Place in IEEE RetroJam 2025 IEEE UP Student Branch	Oct 2025
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- Built a game from scratch in 48 hours, using Rust and Raylib, learning how to **reduce a problem to its core requirements**.

Academic Merit Awards	2020, 2021, 2022
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Participation in RoboCup 2016 RoboCup Federation	Jun 2016
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1st Place in the National Robotics Championship Instituto Politécnico de Bragança	May 2016
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TECHNICAL SKILLS

Programming Languages: Rust, Haskell, C/C++, OCaml, Java, SQL

Frameworks & Tools: Git, UNIX, Tokio, Docker, Plotly, POSIX, STM, gRPC, LibP2P, Dalek, PostgreSQL, Terraform, Raylib and OpenMP

Spoken Languages: Portuguese (Native), English (B2 First, Cambridge), Spanish (Proficient)